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1. Core Functional Modules of the TCB Procurement & Distribution Management System (TPDMS)

1) Procurement Planning Module

- Annual demand planning
- Historical data analysis (based on previous years' consumption)
- Preparation and submission of demand synopsis
- Integration with Estimation Committee for preliminary cost forecasting

2) Tender Management Module

- Tender invitation using OTM/DTM/LTM/RFQ methods
- Tracking of tender document sales through designated bank/accounts section
- Submission and official opening of tenders as per PPR guidelines
- Preparation of Comparative Statement (CS) by the Tender Opening Committee

3) Tender Evaluation & Approval Module

- Formation and operation of the Tender Evaluation Committee
- Verification of submitted tender securities
- Tracking of tender evaluation and approval through appropriate authority (Chairman/Board/Ministry/Cabinet Committee)
- Issuance of Notification of Award (NOA) to selected supplier

4) Contract & LC Management Module

- Collection and verification of performance security from the selected supplier
- Execution of the purchase contract
- Opening of Letter of Credit (LC) for import-based procurement
- Tracking of contract lifecycle and status

5) Supplier & Warehouse Management Module

- Supplier performance monitoring and blacklist/whitelist handling
- Goods receiving, quality control, warehousing, and internal transfers
- Handling of damaged or expired products through financial Right Off process

6) Dealer Management System (DMS)

- Appointment and renewal of dealers based on guidelines
- Field investigation and integration with Deputy Commissioner's office for verification reports
- Collection of dealer security deposit and license fees

7) Distribution Management System

- Distribution of goods from central/regional warehouse to district warehouses and dealers
- Generation of Delivery Orders (DO) and integration with bank payment verification
- Scheduling of truck dispatch and coordination with district administration based on sales calendar

8) Monitoring & Reporting Module

- Monitoring of dealer-wise product sales activities
- Reporting by tag committees at union/upazila/city corporation levels
- Generation of real-time storage, distribution, and contract expiry reports

9) Alerts & Notification System

- Alerts for contract renewal deadlines
- Reminders for delayed product delivery or pending shipments
- Notifications for storage capacity thresholds and warehouse utilization

2. Detail Use Case:

2.1. Use Case Name: Annual Procurement to Contract Execution

Actors:

- Procurement Officer (TCB)
- Estimation Committee
- Tender Committee
- Evaluation Committee
- Chairman/Board/Ministry
- Suppliers/Bidders
- Accounts & Finance Dept.
- Contract Management Unit

Preconditions:

- Annual Procurement Plan is approved.
- Estimated demand has been identified.
- Budget is provisioned.

Basic Flow:

1. **Procurement Officer** prepares a requisition for estimation.
2. **Estimation Committee** determines the estimated price.
3. **Tender Committee** finalizes the tender method (OTM/DTM/etc.) and advertises it via newspaper and website.
4. **Interested Suppliers** purchase tender documents via bank or accounts department.
5. **Suppliers submit tenders** within the given deadline.
6. **Tender Opening Committee** opens tenders and prepares Comparative Statement (CS).
7. **Evaluation Committee** evaluates technical and financial bids.
8. **Approval is taken** from authority based on amount:
 - ≤ 8 crore – Chairman

- ≤ 30 crore – TCB Board
- ≤ 50 crore – Ministry
- 50 crore – Cabinet Committee

9. **Notification of Award (NOA)** is issued to the selected supplier.

10. **Supplier deposits performance security.**

11. **Contract is signed**, and LC (Letter of Credit) is opened (if import-based).

12. **Procurement cycle completes**, and product delivery begins.

Alternative Flows:

- If supplier fails to meet requirements → Tender is rejected.
- If estimated price not approved → Resubmission to Estimation Committee.

Postconditions:

- Contract signed and executed.
- Supplier delivers goods to designated warehouse.

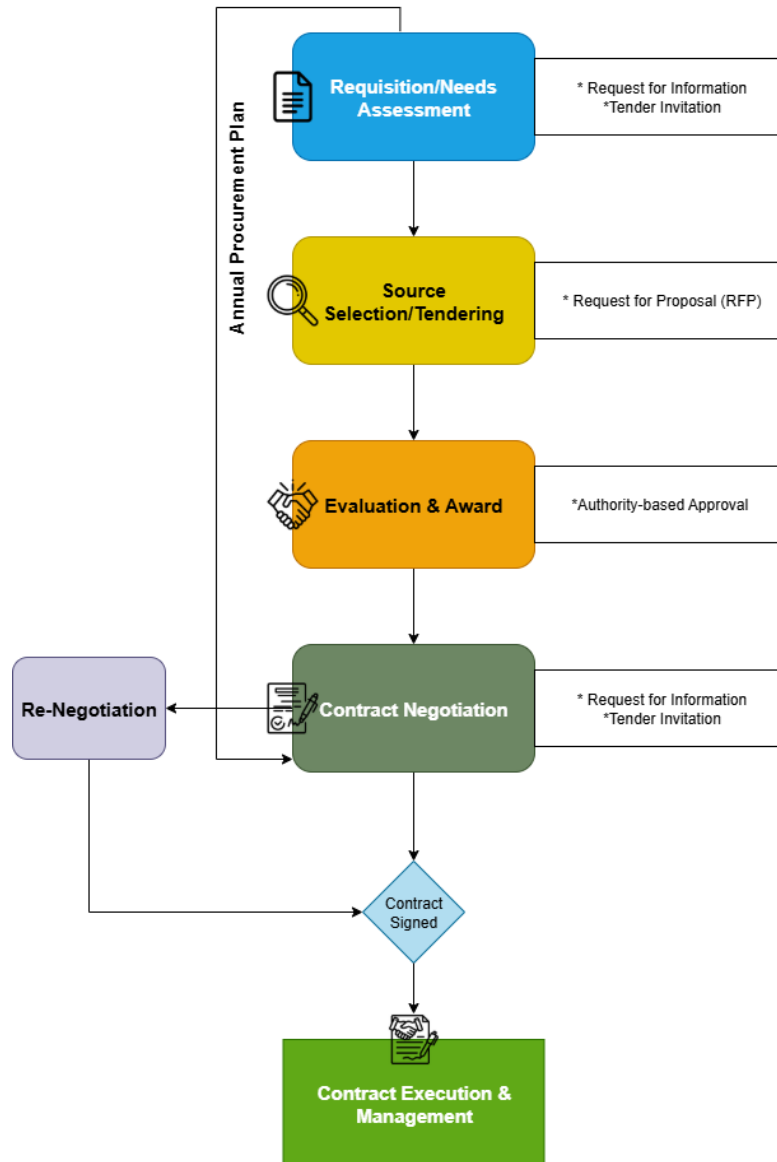


Figure: Annual Procurement to Contract Execution

2.2. Use Case Name: Tender Management Process

Actors:

- Procurement Officer (TCB)
- Tender Committee
- Suppliers/Bidders
- Tender Opening Committee
- Bank/Accounts Department

Preconditions:

- Approved requisition and estimated price are available.
- Procurement method (OTM/DTM/LTM/RFQ) has been selected.
- Procurement notice approval has been obtained.

Basic Flow:

1. **Tender Committee** selects the appropriate tendering method (OTM, DTM, LTM, RFQ) based on procurement rules and the value of the procurement.
2. **Tender notice** is published in designated newspapers and on the TCB website.
3. **Interested Suppliers/Bidders** collect tender documents by paying the specified fee through designated bank or TCB's accounts section.
4. **Suppliers** prepare and submit their sealed tenders within the stipulated deadline.
5. On the **tender opening date**, the **Tender Opening Committee** officially opens all received tenders.
6. The committee prepares a **Comparative Statement (CS)** summarizing all valid bids, including pricing and eligibility.
7. The CS is forwarded to the **Evaluation Committee** for detailed analysis.

Alternative Flows:

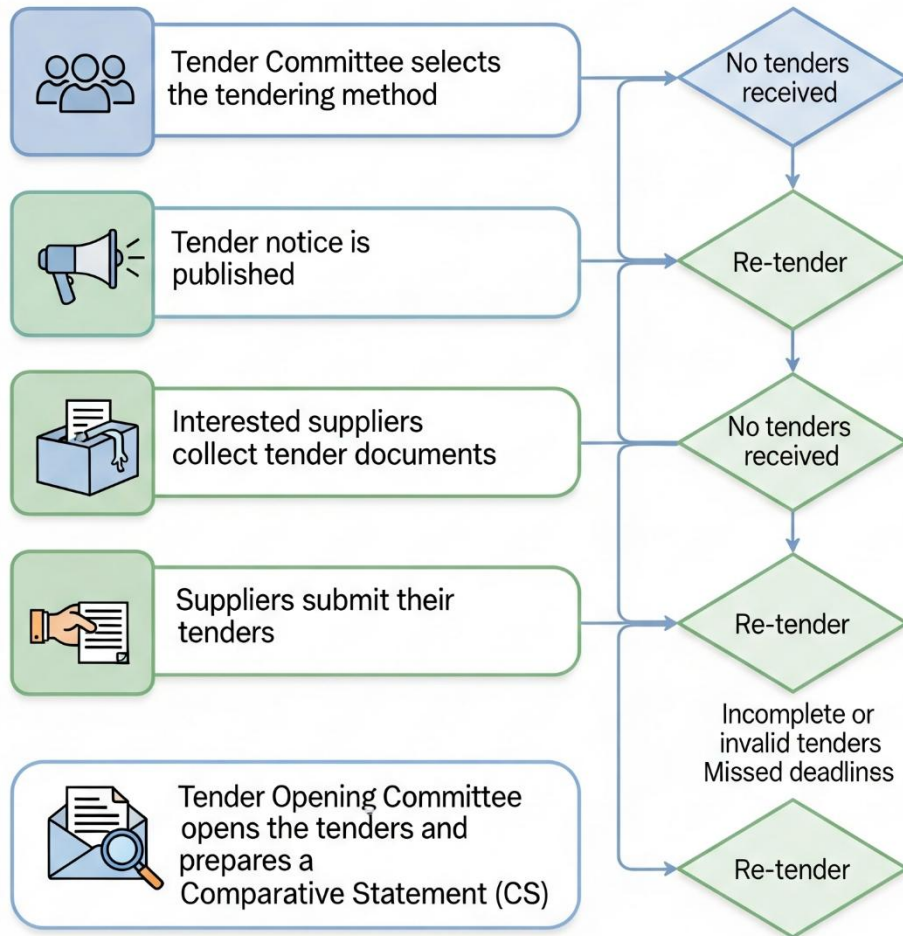
- If no tenders are received → Re-tendering process is initiated.

- If tender documents are incomplete or invalid → Bid is disqualified and marked for exclusion in the CS.
- If the tender deadline is missed → Bid is not accepted as per PPR 2008 guidelines.

Postconditions:

- Valid Comparative Statement (CS) is prepared.
- All eligible tenders are ready for evaluation.
- Bid opening record and documentation are archived.

Tender Management Process



2.3. Use Case Name: Tender Evaluation and Approval Process

Actors:

- Evaluation Committee
- Procurement Officer (TCB)
- Chairman / TCB Board
- Ministry / Cabinet Committee on Public Procurement
- Accounts & Finance Department
- Selected Supplier

Preconditions:

- Comparative Statement (CS) from Tender Opening Committee is available
- All required tender documents are submitted by bidders
- Procurement method and financial thresholds are defined

Basic Flow:

1. **Evaluation Committee** is formed as per PPR 2008 guidelines, comprising technical and financial experts.
2. Committee members review and evaluate all submitted bids, both technical and financial criteria are assessed.
3. **Tender security documents** are verified for validity and compliance.
4. Based on evaluation, the committee prepares a recommendation for award (either acceptance or rejection).
5. The recommendation is sent for **approval based on financial authority level**:
 - ≤ 8 crore → **Chairman**
 - 8 crore & ≤ 30 crore → **TCB Board**
 - 30 crore & ≤ 50 crore → **Ministry**
 - 50 crore → **Cabinet Committee on Public Procurement**
6. Upon receiving the required approval, **Notification of Award (NOA)** is issued to the selected supplier.

7. The **supplier is instructed** to deposit the performance security within the given timeframe.

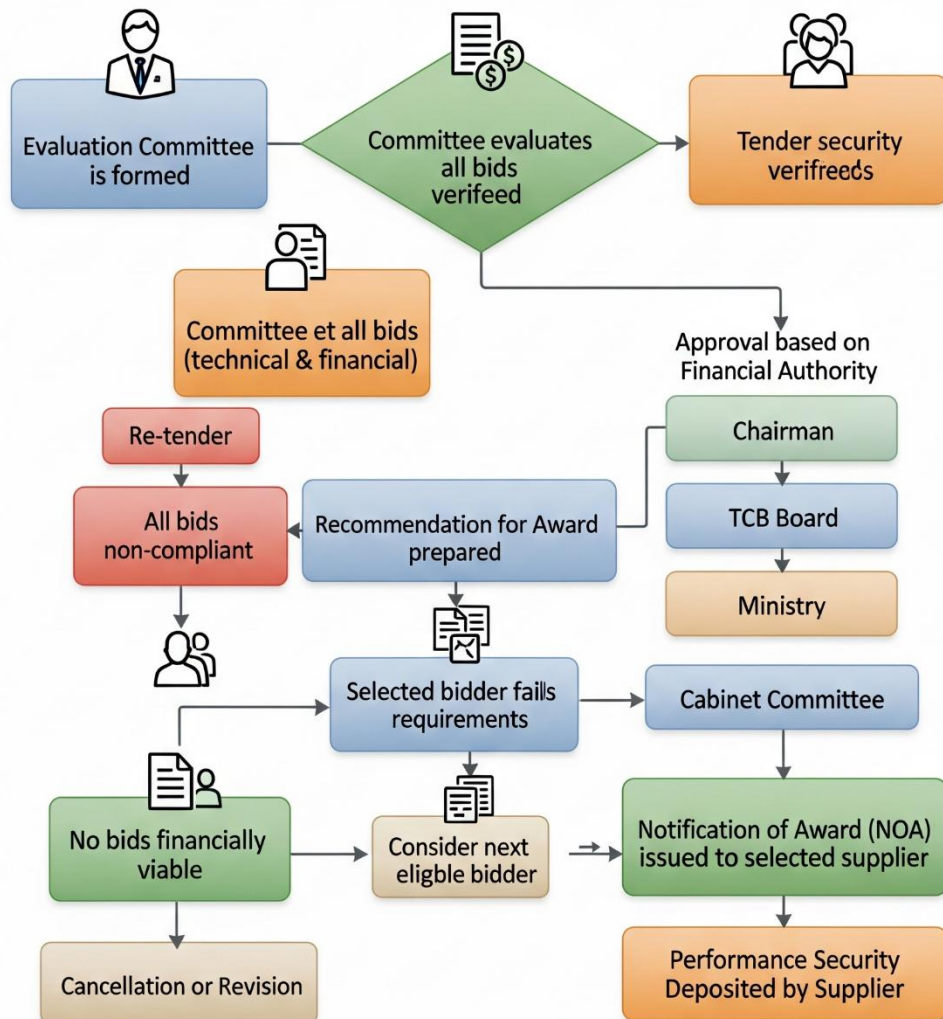
Alternative Flows:

- If evaluation reveals non-compliance in all bids → Re-tender is initiated.
- If the selected bidder fails to meet legal/technical requirements → Next eligible bidder is considered.
- If no bids are financially viable → Tender is canceled or revised.

Postconditions:

- Tender is officially awarded to the selected supplier.
- Performance security is deposited.
- Contract signing phase is initiated.
- All evaluation records are archived and submitted to procurement authority.

Tender Evaluation and Approval Process



2.4. Use Case Name: Contract Execution & LC Management

Actors:

- Selected Supplier
- Procurement Officer (TCB)
- Contract Management Unit
- Accounts & Finance Department
- Bank/LC Issuing Authority

Preconditions:

- Notification of Award (NOA) has been issued to the selected supplier
- Approval from competent authority is received
- Tender evaluation and supplier selection is complete

Basic Flow:

1. **Selected supplier** deposits the required **performance security** (as per NOA terms) through bank or treasury.
2. **Procurement Officer** verifies the deposited performance security and records it in the system.
3. After verification, the **purchase contract** is formally executed between TCB and the supplier, typically on a non-judicial stamp with all terms and conditions included.
4. If the procurement is **import-based**, the **Letter of Credit (LC)** is opened in favor of the supplier via the designated bank, based on contract terms and financial approval.
5. The **Contract Management Unit** tracks all contract-related milestones, including:
 - Delivery deadlines
 - LC maturity
 - Penalties or delays
 - Payment status
6. All updates are logged into the system and relevant stakeholders are notified of important timelines.

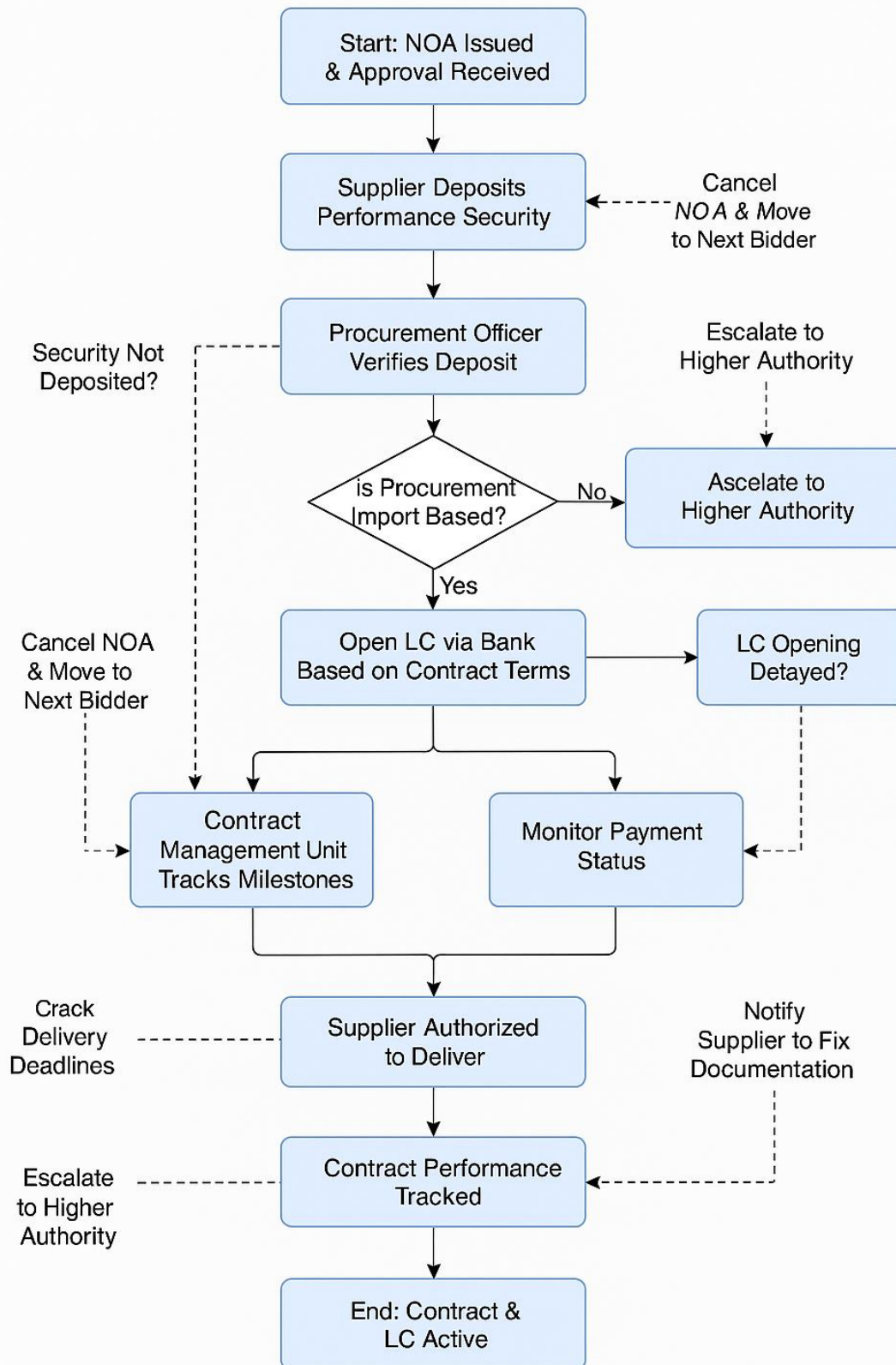
Alternative Flows:

- If supplier fails to deposit performance security → NOA may be cancelled, and next eligible bidder is considered
- If contract execution is delayed → Escalation to higher authority for action
- If LC opening is delayed due to documentation issues → Supplier is notified to rectify

Postconditions:

- Valid contract is signed and archived
- LC is opened (if applicable)
- Supplier is authorized to initiate delivery
- Contract is actively tracked for performance, payment, and delivery compliance

Contract Execution & LC Management



2.5. Use Case Name: Supplier Performance & Warehouse Operations Management

Actors:

- Supplier
- Procurement Officer (TCB)
- Warehouse Manager
- Quality Control Team
- Accounts & Finance Department
- Monitoring Committee

Preconditions:

- Valid contract exists with the supplier
- Delivery schedule is active
- Warehousing facilities are available (owned or rented)

Basic Flow:

1. **Supplier** initiates delivery as per the contract terms.
2. **Warehouse Manager** receives goods at designated TCB warehouse (regional or central).
3. **Quality Control Team** conducts inspection upon arrival using approved checklists and standards (BSTI/BCSI if applicable).
4. Based on inspection:
 - If passed → Goods are stored in the warehouse and entered into the inventory system.
 - If failed → Goods are marked for return or further review.
5. **Supplier performance** (delivery timeliness, compliance, quality) is logged and continuously monitored:
 - Repeated violations may lead to **blacklisting**
 - Consistent compliance may lead to **whitelisting** and fast-track processing
6. If any product is found **damaged or expired** during warehousing:
 - An internal report is prepared by the warehouse/inventory team

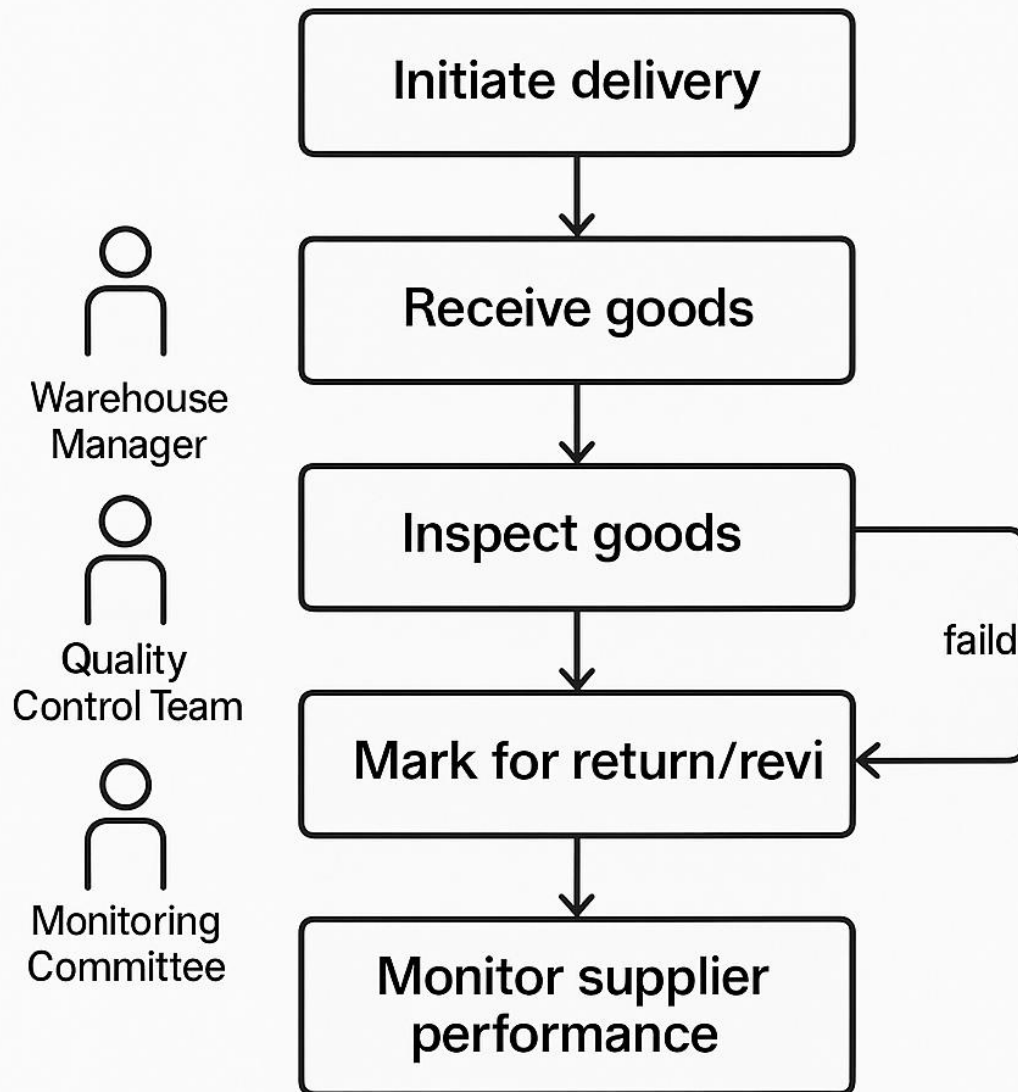
- A committee verifies and assesses the value
 - Based on the report, a **Right Off process** is initiated by the Accounts & Finance Department
7. Inventory is updated accordingly and stakeholders are notified of damage/write-off actions.

Alternative Flows:

- If goods are not delivered within the contract period → Escalation and possible penalty
- If warehouse is full → Rerouting to an alternate facility or holding area
- If quality control fails repeatedly → Procurement team reevaluates supplier eligibility

Postconditions:

- Goods are received, stored, and verified
- Inventory records are updated
- Supplier performance is logged for future evaluation
- Damaged goods are properly accounted and financially written off



2.6. Use Case Name: Dealer Appointment and Management

Actors:

- Dealer Applicant
- Dealer Management Officer (TCB)
- Deputy Commissioner's Office (Field Verification)
- Regional/Field Office
- Accounts & Finance Department

Preconditions:

- A need for dealer recruitment is identified and approved
- Dealer recruitment guidelines (e.g., Guidelines 2021) are in place
- Application window is open and publicly advertised

Basic Flow:

1. **TCB** publishes a **dealer recruitment notice** on its official website and in newspapers.
2. **Interested applicants** submit a formal application with:
 - Non-refundable pay order (BDT 5,000)
 - Required supporting documents (trade license, shop photos, solvency certificate, etc.)
3. All received applications are verified by the **Dealer Management Officer**.
4. **Applications are forwarded** to the **respective Deputy Commissioner's office** for field-level investigation to verify:
 - Authenticity of business
 - Validity of documents (trade license, tenancy agreement, etc.)
5. Upon receiving **positive verification report**, TCB grants **dealer approval**.
6. The approved dealer is directed to:
 - Deposit **security money** (e.g., BDT 30,000)
 - Pay **license fee** (e.g., BDT 6,000)

7. Dealer signs a formal agreement on **non-judicial stamp** (e.g., BDT 300) at the regional/camp office.
8. Every **2 years**, the dealer is required to **renew the dealership** with a renewal fee (e.g., BDT 5,000).

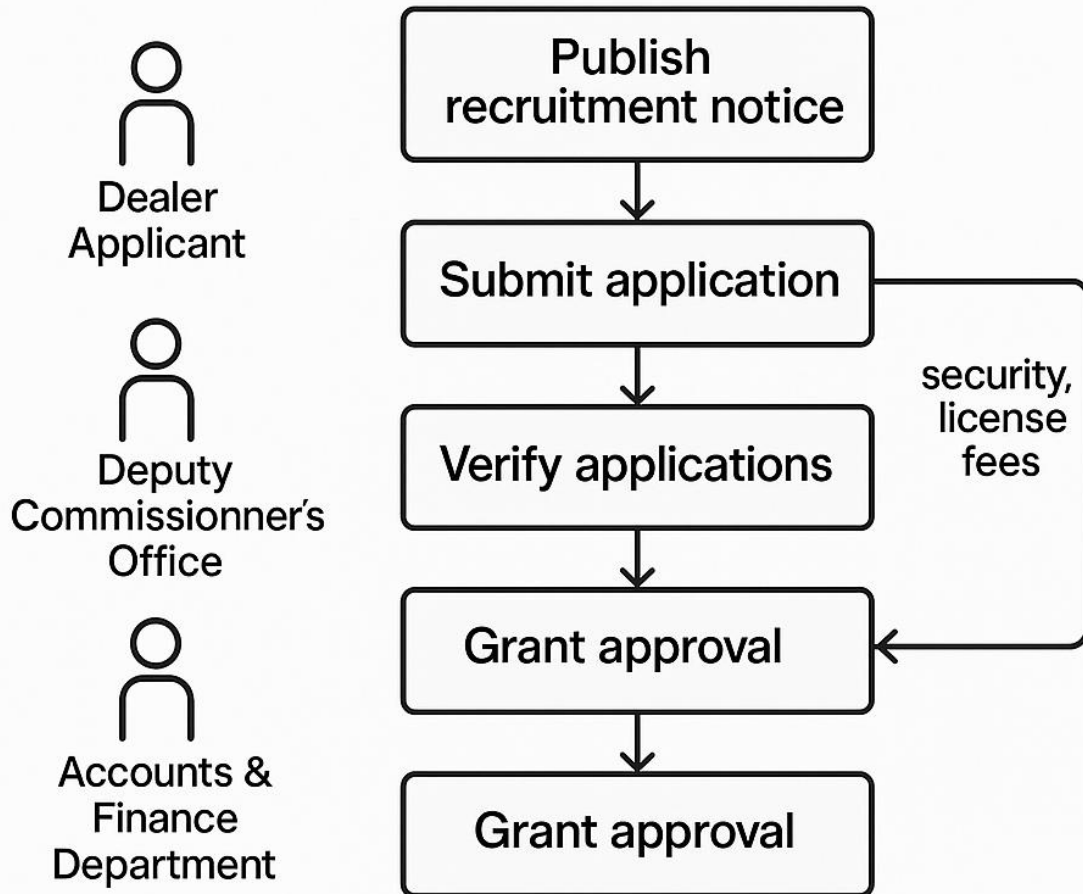
Alternative Flows:

- If field verification fails → Application is rejected and notified
- If security or license fee not paid → Approval remains pending
- If dealer fails to renew on time → Dealership is temporarily suspended or canceled

Postconditions:

- Dealer is officially registered and eligible to receive product allocations
- Security deposit and license fee are recorded in the system
- Dealer profile is maintained and flagged for future renewal tracking

Dealer Appointment and Management



2.7. Use Case Name: Product Distribution and Delivery Order Management

Actors:

- Distribution Officer (TCB)
- Warehouse Manager (Central, Regional, District)
- Approved Dealer
- District Administration
- Bank (for payment verification)
- Transport Contractor / Logistics Team

Preconditions:

- Products are available at TCB central or regional warehouse
- Dealer allocation has been approved
- Sales calendar is prepared in consultation with the District Administration
- Bank account is designated for payment by dealers

Basic Flow:

1. **Distribution Officer** prepares a list of dealers and their allocated quantities based on approved quota.
2. **Delivery Order (DO)** is generated for each dealer.
3. Dealer deposits the **payment to TCB's designated bank account** at least one day prior to collection.
4. Bank confirms payment, and the system automatically **verifies the deposit**.
5. Upon verification, DO is activated and shared with the **warehouse manager**.
6. **Warehouse Manager** schedules **product release** for the dealer.
7. Based on the **sales calendar**, trucks are dispatched:
 - From central/regional warehouse to district warehouse
 - Or directly to dealers, as instructed by district administration
8. **Dealer collects goods** at their own expense, presenting DO and payment slip.

9. Distribution record is updated in the system with delivery time, quantity, and receiving acknowledgment.
10. Relevant stakeholders are notified and daily movement summary is archived.

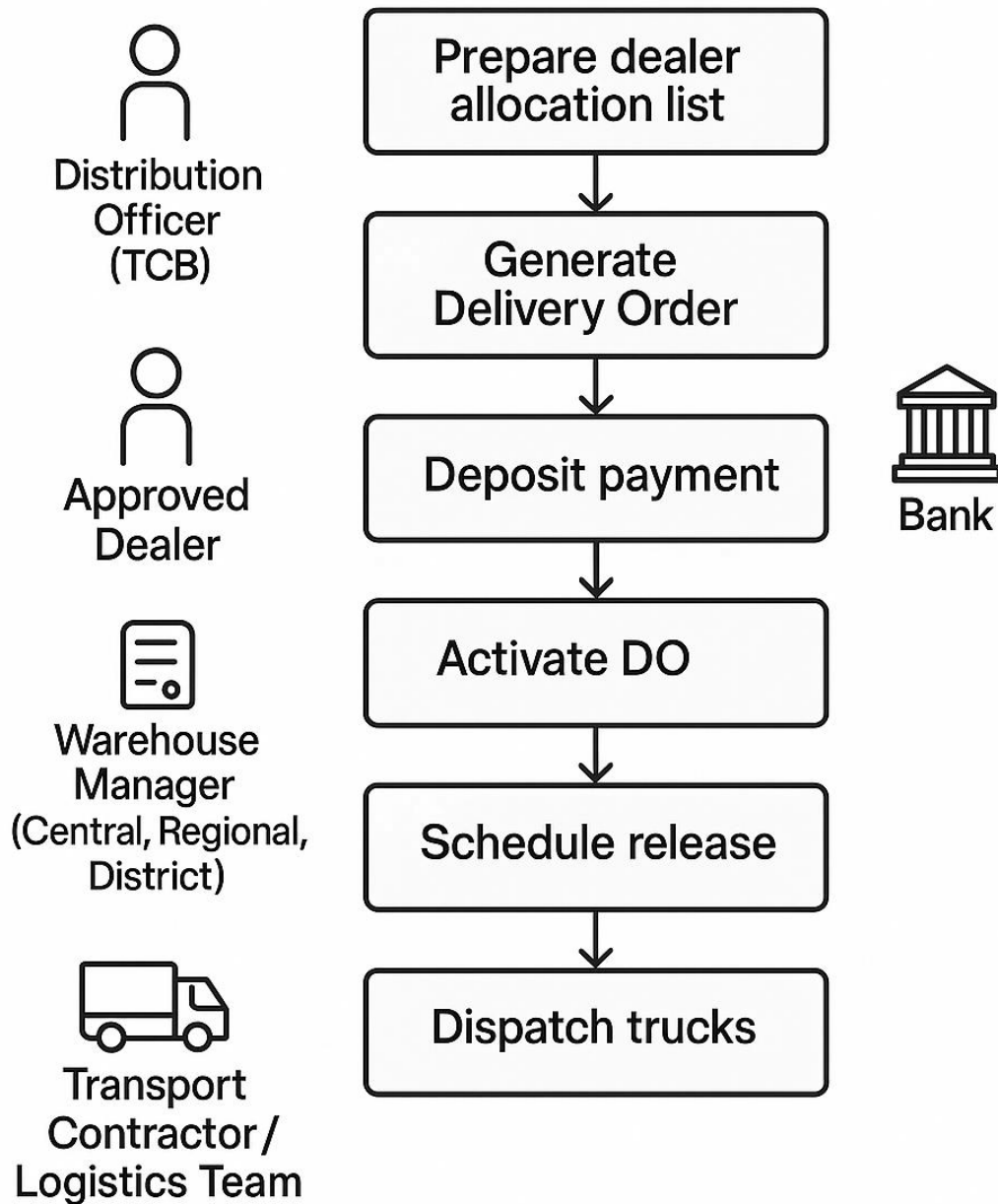
Alternative Flows:

- If payment is not verified → DO remains inactive, and dealer cannot collect goods
- If truck dispatch is delayed → System escalates alert to logistics team and district administration
- If warehouse stock is insufficient → Distribution Officer reschedules delivery or reallocates

Postconditions:

- Products are successfully delivered to the dealer or district warehouse
- Delivery confirmation is logged
- Financial reconciliation is updated in real-time
- System generates delivery and movement report for daily monitoring

Product Distribution and Delivery Order Management



2.8. Use Case Name: Sales Monitoring and Reporting

Actors:

- Monitoring Officer (TCB Head/Regional Office)
- Dealer
- Tag Committee (Union/Upazila/City Corporation level)
- District Administration
- System Admin / MIS Unit

Preconditions:

- Product distribution has started
- Dealers are actively selling products to beneficiaries
- Tag Committees are formed as per government order
- Dealers are required to report sales with supporting documentation

Basic Flow:

1. **Dealers** begin product sales to beneficiaries based on government-approved sales calendar.
2. After each sales cycle/day, dealers **submit sales data** through the system—this may include:
 - Quantity sold
 - Number of beneficiaries served
 - Geo-location (if GPS-enabled system exists)
3. **Monitoring Officers** (TCB regional/central) log into the dashboard to view dealer-wise sales trends and flags.
4. **Tag Committees** visit distribution points physically and verify:
 - Whether actual sales took place
 - Whether beneficiaries presented valid family cards
 - Whether dealer followed pricing and quantity limits
5. Committees submit **verification reports** using system-defined forms (e.g., Appendix-01A, 01B, 01C).
6. The system **generates automated analytics and reports**, including:

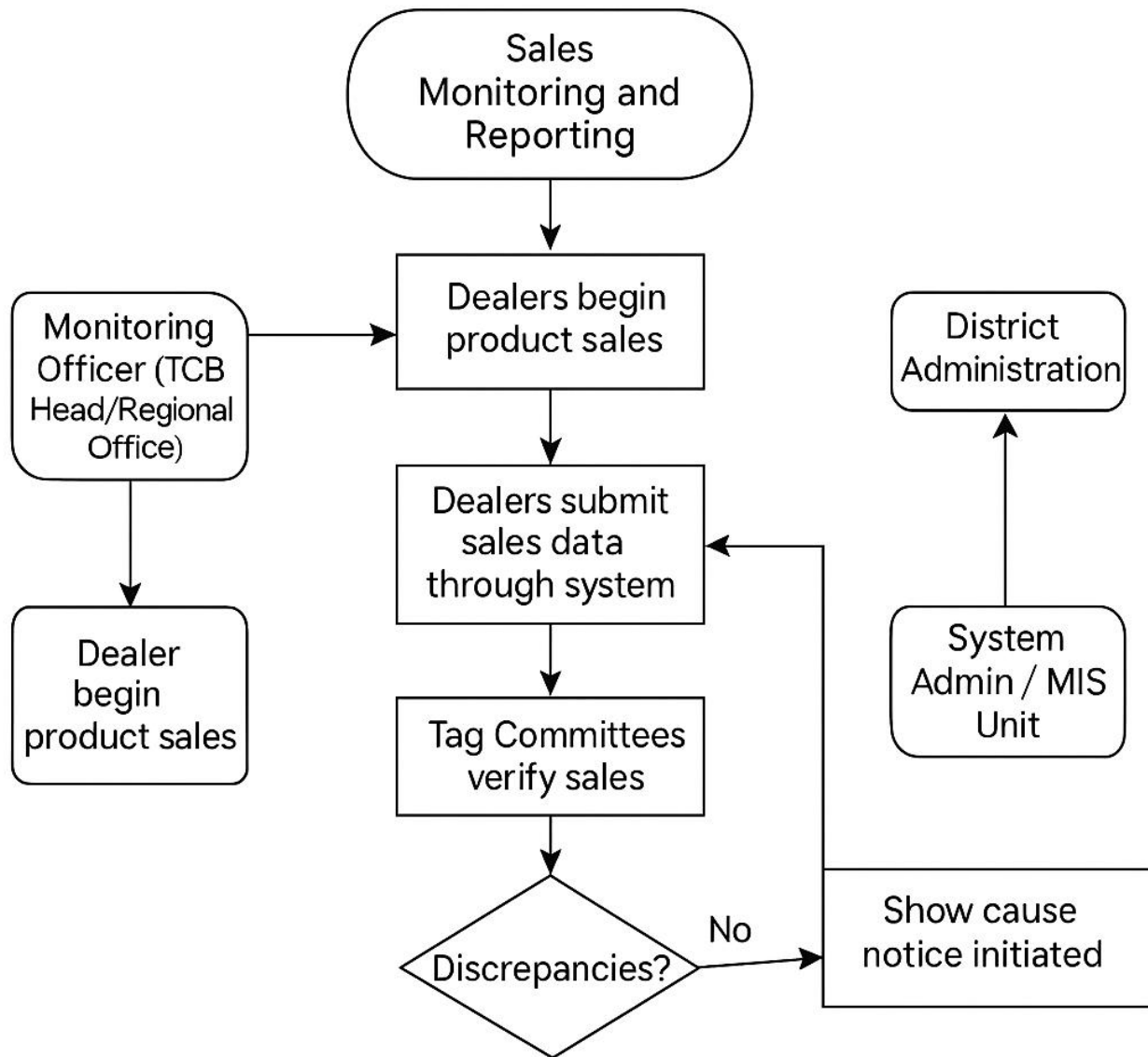
- Dealer compliance status
 - Stock-out alerts
 - Unsold product inventory
7. In parallel, the system also generates:
- **Warehouse storage reports** (opening, received, closing balance)
 - **Distribution reports** (warehouse to dealer, or inter-district transfer)
 - **Contract expiry alerts** for supplier and dealer agreements
8. MIS/Admin users download reports and send summaries to higher management or Ministry.

Alternative Flows:

- If dealer fails to submit sales data → Escalation alert is triggered
- If Tag Committee finds discrepancies → Show cause notice initiated
- If reports mismatch → Audit team is engaged for field verification

Postconditions:

- Sales activities are properly monitored
- Field-level validations are recorded and archived
- Real-time inventory and operational data is available for management decision-making
- Expiring contracts are proactively flagged for renewal



System generates analytics and reports

2.9. Use Case Name: Automated Alerts & Notification Management

Actors:

- Procurement Officer
- Contract Management Unit
- Warehouse Manager
- Dealer/ Supplier
- System Admin / Automation Engine

Preconditions:

- Active contracts and delivery schedules are available in the system
- Delivery dates, renewal dates, and storage thresholds are configured
- Notification rules and triggers are set in the system

Basic Flow:

1. The system continuously monitors key data points:
 - Contract expiry dates
 - Product delivery schedules
 - Warehouse storage capacity utilization
2. When a **contract renewal date** is approaching (e.g., 30, 15, 7 days before), the system sends:
 - **Email/SMS/Push notifications** to responsible officers
 - Reminders to dealers/suppliers for contract extension
3. If a **product delivery is delayed**, the system identifies it based on the contract timeline and:
 - Sends alerts to the Procurement Officer and Contract Manager
 - Notifies the supplier/dealer with escalation if required
4. The system calculates **storage capacity usage** for each warehouse daily.
 - If capacity crosses a defined threshold (e.g., 80%, 90%, 100%), alerts are sent to:

- Warehouse Manager
 - Central Inventory Controller
 - Regional Heads
5. Notification dashboard shows all **pending actions**, and logs alert history for audit.

Alternative Flows:

- If alert delivery fails (e.g., email server down) → Notification is retried and logged
- If warehouse data is outdated → System prompts warehouse staff for daily entry

Postconditions:

- All relevant users are informed on time to take proactive actions
- Delays, bottlenecks, and overutilization are prevented
- Compliance and operational efficiency improve through timely responses

3. Dashboard

SL	Component	Description	Visualization	Target User
1	Procurement Snapshot	মোট চলমান টেন্ডার, মূল্যায়নধীন টেন্ডার, এবং স্বাক্ষরিত চুক্তির সংখ্যা।	KPI Cards	Management, Procurement Officer
2	Financial Overview	চলতি অর্থবছরে মোট চুক্তির মূল্য, বাজেট বনাম খরচ, এবং মোট পারফরম্যান্স সিকিউরিটির পরিমাণ।	Pie Chart / Bar Chart	Management, Accounts Dept.
3	Inventory At a Glance	কেন্দ্রীয় ও বিভাগীয় ওয়ারহাউজে প্রধান পণ্যগুলোর (তেল, চিনি, ডাল ইত্যাদি) মোট মজুদের পরিমাণ।	Donut Chart / Number Cards	Management, Warehouse Manager
5	Warehouse Capacity	প্রতিটি ওয়ারহাউজের বর্তমান ধারণক্ষমতার কত শতাংশ ব্যবহৃত হচ্ছে।	Gauge Chart / Progress Bar	Warehouse Manager, Distribution Officer
6	Distribution Summary	চলতি মাসে মোট বিতরণ করা পণ্যের পরিমাণ, মোট ইস্যু করা ডেলিভারি অর্ডার (DO) এবং মোট বিক্রির পরিমাণ (টাকায়)।	Line Chart (Last 30 day's Trend)	Management, Distribution Officer
7	Dealer Status	মোট সক্রিয় ডিলার, নতুন অনুমোদিত ডিলার এবং নবায়নের অপেক্ষায় থাকা ডিলারের সংখ্যা।	KPI Cards	Dealer Management Officer
8	Pending Approvals	আর্থিক ক্ষমতা অনুযায়ী অনুমোদনের জন্য অপেক্ষমাণ ফাইলের তালিকা (যেমন: চেয়ারম্যানের জন্য, বোর্ডের জন্য)।	List / Table	Chairman, Board Members
9	Critical Alerts	সিস্টেম থেকে তৈরি হওয়া জরুরি নোটিফিকেশন, যেমন: চুক্তির মেয়াদ শেষ হতে চলেছে, ডেলিভারি বিলম্বিত, ওয়ারহাউজ পূর্ণ।	Notification Panel	All relevant users
10	Sales Monitoring Map	বাংলাদেশের একটি ম্যাপে বিভাগ/জেলা অনুযায়ী বিক্রির পরিমাণ বা ডিলারের অবস্থান দেখানো হবে।	Geographic Map	Management, Monitoring Officer

4. Reports

1. Procurement & Tender Report
2. Evaluation & Approval Report
3. Contract & LC Management Report
4. Supplier Performance Report
5. Warehouse & Inventory Report
6. Dealer Management Report
7. Distribution & Sales Report
8. Financial Reconciliation Report